

ALL INDIA COUNCIL FOR TECHNICAL EDUCATION

Ministry of Education • Government of India

FULL STACK WEB DEVELOPMENT INTEGRATION INITIATIVE

YEAR OF AI • 2025

Empowering India's Higher & Technical Education Ecosystem

40 M+

Students Targeted

14,000+

Institutions Covered

2021

LITE Programme Launched

AICTE formally declares 2025 as the **Year of Full Stack Web Development** — embedding Full Stack Web Development across India's higher and technical education ecosystem of **14,000+ institutions** and **40 million students**.

EXECUTIVE SUMMARY

The All India Council for Technical Education (AICTE), apex statutory body under the Ministry of Education, Government of India, has adopted a comprehensive national strategy to embed **Full Stack Web Development (FSWD)** as a core competency across all disciplines of higher and technical education.

The initiative is structured under the **AICTE LITE Programme** in partnership with **Pupilfirst** and the **Global Developer Corps (GDC) 2024–2029**, targeting 2.4 crore students and 1,00,000 faculty over six years.

This initiative aligns with the **National Education Policy (NEP) 2020**, Skill India, Digital India, and the vision of **Viksit Bharat 2047**. Through industry partnerships, FSWD-integrated curricula, and large-scale faculty development, AICTE aims to position India as a globally competitive hub for full-stack engineering talent.

INSTITUTIONAL WEB DEVELOPMENT PLEDGE

Each AICTE-approved institution is formally requested to adopt the following commitment:

"We pledge to position India as a Vishwa Guru in web technology — leading global innovation, upholding ethical standards, and advancing Full Stack Web Development education in fulfilment of the Hon'ble Prime Minister's vision."

This pledge commits each institution to: integrate FSWD across all disciplines; foster applied projects and startup incubation; prepare graduates for industry-ready full-stack roles; and champion ethical, accessible web development. Institutions must hold formal adoption ceremonies and display the pledge prominently on campus and in communications.

STRATEGIC PARTNERSHIP: AICTE LITE — PUPILFIRST & GLOBAL DEVELOPER CORPS (GDC) 2024–2029

Pupilfirst has confirmed its commitment to deliver industry-aligned Full Stack Web Development education to students across AICTE-approved institutions. The partnership operates under the **Global Developer Corps (GDC) Artificial Intelligence Workforce Scheme 2024–2029** and is structured as a phased deployment model ensuring sustainable adoption and measurable impact.

Phase 1	Phase 2	Phase 3
Digital Outreach & Awareness	Faculty Enablement & Pilot Deployment	Institutional Student Onboarding
Mass digital distribution of FSWD learning resources through AICTE portals, official circulars, and institutional mailer networks. Baseline awareness of WD101 (HTML, CSS, JavaScript) established across all 14,000+ institutions.	Structured faculty training sessions integrating WD201 (Node.js, Express.js) and WD301 (React & TypeScript) into teaching and research workflows. Pilot launches within Centres of Excellence generate evidence for full-scale rollout.	Full-scale student access via web and mobile through formal institutional onboarding. Access as Minor Degree (18–20 credits) or Open Elective via Academic Bank of Credits (ABC), aligned with NEP 2020's emphasis on experiential and inquiry-based learning.

ACTION PLAN: FIVE STRATEGIC PILLARS

#	PILLAR	KEY ACTIVITIES
I	FSWD Affirmation Pledge	Formal institutional adoption ceremonies; campus-wide display; public accountability integration.
II	Campus-Wide Web Dev Campaign	Web Dev Awareness Week; student Developer Chapters; industry-aligned Web Labs; Career Pathways counselling.
III	FSWD-First Curriculum Redesign	Foundational and advanced FSWD modules at UG/PG level; interdisciplinary integration for non-CS branches.
IV	Faculty Development	ATAL Academy FDPs; PG certificate programmes; collaborations with Pupilfirst, Microsoft, AWS, IITs, INAE.
V	Industry Collaboration	Internships, sponsored R&D projects, mentorships, hackathons co-organised with MoE-MIC and GDC partners.

FSWD-INTEGRATED CURRICULUM

AICTE-Approved Minor Degree in Advanced Web Development (WD101–WD401) provides an **18–20 credit framework** spread over 2–3 semesters. Students from any branch who have completed their second semester may enrol. Curriculum is dynamically maintained by industry experts and updated with AICTE under the GDC Scheme 2024–2029.

COURSE	LEVEL	TITLE	KEY TOPICS COVERED	CR
WD 101	Foundation	Getting Started with JavaScript	HTML5 & Semantic Markup; CSS3 & Tailwind CSS; JavaScript fundamentals; DOM manipulation; Event handling. <i>Capstone: Static web application</i>	2
WD 201	Intermediate	Server-Side Programming	Node.js & Express.js; RESTful API design; PostgreSQL / MySQL; User authentication & session management. <i>Capstone: Full-stack CRUD application</i>	6
WD 301	Advanced	Front-End Development with React & TypeScript	React.js component architecture; TypeScript; Redux state management; React Router; API integration; Responsive UI design. <i>Capstone: Single Page Application (SPA)</i>	6
WD 401	Production	Getting Ready for Production	Testing (Jest); CI/CD with GitHub Actions; Docker & containerisation; Cloud deployment (Render / AWS); Error monitoring (Sentry); GenAI application foundations. <i>Capstone: Industry-grade deployment</i>	6

TOTAL PROGRAMME CREDITS: 20 | Duration: 2–3 Semesters | Eligible: Any Branch, Post 2nd Semester | Delivery: Learner-Centred (NEP 2020)

FACULTY DEVELOPMENT (ATAL ACADEMY & LITE PROGRAMME)

The AICTE Training and Learning (ATAL) Academy, in collaboration with LITE partner institutions, drives large-scale faculty upskilling through structured FDPs in Full Stack Web Development, DevOps, and emerging web technologies. Key programmes conducted during 2023–24:

Programme	Partners	Focus Areas	Trained
LITE FSWD FDP	Pupilfirst / AICTE	WD101–WD401, Learner-Centred Pedagogy, NEP 2020 integration	50+ Institutions
Tech Saksham FDP	Microsoft & SAP (Joint CSR)	Cloud Computing, Web APIs, IR 4.0 Technologies, Azure Dev	4,516
Next Gen Employability	Edunet / EY Global Delivery	MERN Stack, Industry 4.0, Enterprise Web Technologies	3,556
GDC AI Workforce Scheme	Pupilfirst / Global Developer Corps	GenAI, CI/CD, Docker, Production-Grade Web Dev	Multi-batch
PG Certificate Programme	AICTE / ATAL Academy	Web Architecture, API Design, Cloud, Security & DevOps	1,000+

WEB DEV LABS & INFRASTRUCTURE

Each AICTE-approved institution is encouraged to establish a **Full Stack Web Development Lab** as part of its Industry Innovation Council (IIC). The recommended facility encompasses:

- High-speed workstations with Node.js, React & DevOps toolchains
- Cloud sandbox environments (AWS / Azure credits via NEAT partners)
- CI/CD pipeline rigs — Docker and GitHub Actions demonstrations
- Open-source incubation bays for student-led project development
- UX & accessibility testing stations for responsive web design
- AI-assisted code review tools aligned with GDC 2024–2029 Scheme

INDUSTRY COLLABORATION

AICTE's industry engagement model systematically closes the skill gap between academic output and industry demand. Key mechanisms operationalised across approved institutions:

- **Sponsored R&D Projects** — Industry partners define real-world problem statements; student and faculty teams deliver full-stack solutions under structured mentorship.
- **Internship & Pre-Placement Pipelines** — Formalised arrangements with tech companies providing hands-on exposure to production-grade web systems.
- **MoE-MIC Hackathons** — Co-organised competitions stimulating collaborative full-stack engineering and entrepreneurial ideation.
- **Guest Faculty & Practitioners** — Industry engineers embedded as adjunct faculty and curriculum consultants.
- **Free Developer Tools** — GitHub, VS Code, cloud credits (AWS/Azure) and Pupilfirst platform access democratised across all institutions.

Leadership: Prof. T.G. Sitharam, Chairman AICTE, has directed Full Stack Web Development to be recognised as a critical employability skill under **The Road Ahead 2.0** — the strategic roadmap for India's technology education transition. The LITE programme is mandated to train **2.4 crore students** and **1,00,000 faculty** over the next six years, creating a continuous pipeline of industry-ready full-stack developers.

POLICY ALIGNMENT & NATIONAL VISION

AICTE's Full Stack Web Development integration framework is architecturally aligned with India's flagship national policy programmes and the long-term developmental vision of **Viksit Bharat 2047**:

NATIONAL PROGRAMME	ALIGNMENT WITH AICTE FSWD INITIATIVE
NEP 2020	Experiential and inquiry-based FSWD pedagogy operationalising critical thinking, project-based learning, and real-world deployment via Minor Degree framework and Academic Bank of Credits (ABC).
Skill India	FSWD skilling programmes targeting employability in the digital economy — full-stack engineers, UI/UX developers, and cloud-native application specialists.
Digital India	FSWD-powered institutional platforms, digital-first administrative tools, and student-led web applications advancing India's digital public infrastructure across 14,000+ institutions.
Make in India	Incubation support, innovation hackathons, and industry co-creation fostering indigenous web product development and deep-tech entrepreneurship.
Viksit Bharat 2047	Building long-term web technology human capital — full-stack engineers, DevOps practitioners, and digital entrepreneurs sustaining India's global competitiveness through 2047 and beyond.

MANDATORY COMPLIANCE & INSTITUTIONAL OBLIGATIONS

- All AICTE-approved institutions offering CSE, IT, or BCA programmes must designate a **FSWD Nodal Coordinator** by 31st March 2025.
- Institutions are required to integrate at least one FSWD elective (WD101 / WD201 equivalent) as an Open Elective in UG programmes by **AY 2025–26**.
- Faculty members must complete a minimum of **one ATAL Academy-certified FSWD FDP** within the academic year.
- Institutions shall submit an annual **FSWD Implementation Report** to AICTE through the approval portal, documenting enrolments, completions, and placement outcomes.
- Industry Innovation Councils (IICs) must co-host at least **one web development hackathon or project showcase** per semester in collaboration with industry partners.

AICTE invites all stakeholders — institutions, faculty, students, industry partners, and policymakers — to actively participate in this transformative national journey. Together, we shall make India a global hub for Full Stack Web Development talent, digital innovation, and human excellence — fulfilling the aspirations of **Viksit Bharat 2047**.